

Theory and Practical Implementation of VQE for Molecular Systems

A Hybrid Quantum-Classical Algorithm

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Why Did VQE Emerge?

Central Goal: Solve the ground-state eigenvalue problem

$$\hat{H}|\psi_0\rangle = E_0|\psi_0\rangle \quad (1)$$

The Problem with Classical Methods

- Hilbert space grows exponentially with system size
- Exact diagonalization becomes infeasible
- Quantum chemistry is naturally quantum mechanical

Early Quantum Approach: Quantum Phase Estimation (QPE)

- Polynomial-time quantum algorithm
- Requires deep circuits and long coherence times
- Impractical for noisy NISQ devices

Key Idea Behind VQE

- Replace one long coherent evolution
- With many short quantum measurements
- Hybrid quantum-classical optimization loop

$$E(\theta) = \langle \psi(\theta) | \hat{H} | \psi(\theta) \rangle \geq E_0$$

(2)

Hilbert Space

Griffiths once said "All quantum states live in the **Hilbert Space**".

A quantum state is represented using a **Ket Vector**

$$|\psi\rangle = \begin{bmatrix} c_0 \\ c_1 \\ \vdots \\ \end{bmatrix} \quad (3)$$

where the entries are generally complex numbers, which obey the rule $|c_0|^2 + |c_1|^2 + \dots = 1$.

The corresponding **Bra Vector** is obtained by taking the transpose and complex conjugate:

$$\langle\psi| = [c_0^* \quad c_1^* \quad \dots] \quad (4)$$

Quantum states must always be normalized:

$$\langle\psi|\psi\rangle = [c_0^* \quad c_1^* \quad \dots] \begin{bmatrix} c_0 \\ c_1 \\ \vdots \\ \end{bmatrix} = |c_0|^2 + |c_1|^2 + \dots = 1 \quad (5)$$

A quantum gate or operator acts as a transformation on the state-vector:

$$\hat{O} |\psi\rangle \tag{6}$$

The expectation value of an operator is

$$\langle\psi| \hat{O} |\psi\rangle \tag{7}$$

which can be roughly interpreted as:

“I applied the operation $\hat{O}$ to the state $|\psi\rangle$. How much does the transformed state resemble the original state?”

In particular, when the operator is the Hamiltonian $\hat{H}$,

$$\boxed{\langle\psi| \hat{H} |\psi\rangle} \tag{8}$$

gives the energy expectation value of the quantum state.

Variational Principle : Intuition

Suppose we have a Hamiltonian represented by the matrix

$$H = \begin{bmatrix} a & b \\ b^* & d \end{bmatrix} \quad (9)$$

Our goal is to find a normalized vector

$$|\psi\rangle = \begin{bmatrix} p \\ q \end{bmatrix}, \quad |p|^2 + |q|^2 = 1 \quad (10)$$

such that the quantity (**expectation value of energy**)

$$E(\psi) = \langle \psi | H | \psi \rangle = [p^* \quad q^*] \begin{bmatrix} a & b \\ b^* & d \end{bmatrix} \begin{bmatrix} p \\ q \end{bmatrix} \quad (11)$$

becomes as small as possible.

In other words, minimize $\frac{\langle \psi | H | \psi \rangle}{\langle \psi | \psi \rangle}$ **by choosing an appropriate** $|\psi\rangle$ **and placing appropriate parametrization on it.**
Experimentally and mathematically, we observe:

The minimum possible value occurs when $|\psi\rangle$ is the eigenvector corresponding to the lowest eigenvalue of H .

For a general N -dimensional Hamiltonian:

$$E_{\min} = E_0 \quad (12)$$

where E_0 is the ground-state eigenvalue.

But why does minimizing $\langle \psi | H | \psi \rangle$ always lead us to the ground state?

Variational Principle (Contd.)

Consider a 2×2 Hamiltonian matrix

$$H = \begin{bmatrix} E_0 & 0 \\ 0 & E_1 \end{bmatrix}, \quad E_0 < E_1 \quad (13)$$

The eigenvectors of H are

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad (14)$$

which form a complete basis. We use them to construct a trial state.

Now consider a normalized trial vector

$$\psi = c_0 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + c_1 \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} c_0 \\ c_1 \end{bmatrix}, \quad |c_0|^2 + |c_1|^2 = 1 \quad (15)$$

The expectation value of the energy is

$$E(\psi) = \begin{bmatrix} c_0^* & c_1^* \end{bmatrix} \begin{bmatrix} E_0 & 0 \\ 0 & E_1 \end{bmatrix} \begin{bmatrix} c_0 \\ c_1 \end{bmatrix} = |c_0|^2 E_0 + |c_1|^2 E_1 \quad (16)$$

The energy expectation value is therefore a weighted average of the eigenvalues.

Key Result

The minimum possible expectation value is

$$E(\psi) \geq E_0 \quad (17)$$

Equality occurs only when

$$c_1 = 0 \text{ and } c_0 = 1 \quad (18)$$

which means the vector becomes

$$|\psi\rangle = c_0 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + c_1 \begin{bmatrix} 0 \\ 1 \end{bmatrix} = (1) \begin{bmatrix} 1 \\ 0 \end{bmatrix} + (0) \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad (19)$$

i.e., the eigenvector corresponding to the lowest eigenvalue E_0 .

- **This principle, allows us to give an upper bound to the true ground state energy, by selecting an appropriate $|\psi\rangle$.**
- *Of course, the upper bound is closer to the ground state energy, the better our guess of $|\psi\rangle$ is.*

A Simple Variational Problem

Suppose we are given the Hamiltonian

$$H = 0.6X + 0.7Z \quad (20)$$

Using the Pauli matrices

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \quad \Rightarrow \quad H = \begin{bmatrix} 0.7 & 0.6 \\ 0.6 & -0.7 \end{bmatrix} \quad (21)$$

- Can we devise a method to solve this Hamiltonian?
- More specifically:

Can we find a state-vector $\begin{bmatrix} a \\ b \end{bmatrix}$ which gives the lowest possible energy expectation value?

The expectation value of the Hamiltonian is

$$E(a, b) = \begin{bmatrix} a^* & b^* \end{bmatrix} \begin{bmatrix} 0.7 & 0.6 \\ 0.6 & -0.7 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} \quad (22)$$

subject to the normalization condition

$$|a|^2 + |b|^2 = 1 \quad (23)$$

Goal

Find the values of a and b that minimize $E(a, b)$.

A Simple Variational Problem (Contd.,)

- Let's use Quantum Computing to approach the problem.
- We use a gate $R_y(\theta) = \begin{bmatrix} \cos\left(\frac{\theta}{2}\right) & -\sin\left(\frac{\theta}{2}\right) \\ \sin\left(\frac{\theta}{2}\right) & \cos\left(\frac{\theta}{2}\right) \end{bmatrix}$ to parametrize the initial state $|0\rangle$.
- Our circuit looks like this :

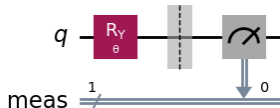


Figure: A guess circuit for finding the solution of $H = 0.6 X + 0.7 Z$

Born-Oppenheimer Approximation

$$\hat{H}_{\text{mol}} = \underbrace{-\sum_{A=1}^{N_n} \frac{1}{2M_A} \nabla_A^2}_{\text{kinetic (nucleon)}} - \underbrace{\sum_{i=1}^{N_e} \frac{1}{2} \nabla_i^2}_{\text{kinetic (electron)}} - \underbrace{\sum_{i=1}^{N_e} \sum_{A=1}^{N_n} \frac{Z_A}{|\mathbf{r}_{iA}|}}_{\text{electron-nucleon attraction}} + \underbrace{\frac{1}{2} \sum_{\substack{i,j \\ i \neq j}}^{N_e} \frac{1}{|\mathbf{r}_{ij}|}}_{\text{electron-electron repulsion}} + \underbrace{\frac{1}{2} \sum_{\substack{A,B \\ A \neq B}}^{N_n} \frac{Z_A Z_B}{|\mathbf{R}_{AB}|}}_{\text{nucleon-nucleon repulsion}}$$

$$\hat{H}_{\text{elec}} = \cancel{\underbrace{-\sum_{A=1}^{N_n} \frac{1}{2M_A} \nabla_A^2}_{\text{kinetic (nucleon)}}} - \underbrace{\sum_{i=1}^{N_e} \frac{1}{2} \nabla_i^2}_{\text{kinetic (electron)}} - \underbrace{\sum_{i=1}^{N_e} \sum_{A=1}^{N_n} \frac{Z_A}{|\mathbf{r}_{iA}|}}_{\text{electron-nucleon attraction}} + \underbrace{\frac{1}{2} \sum_{\substack{i,j \\ i \neq j}}^{N_e} \frac{1}{|\mathbf{r}_{ij}|}}_{\text{electron-electron repulsion}} + \cancel{\underbrace{\frac{1}{2} \sum_{\substack{A,B \\ A \neq B}}^{N_n} \frac{Z_A Z_B}{|\mathbf{R}_{AB}|}}_{\text{nucleon-nucleon repulsion}}}$$

$V_{NN}(\text{constant})$

Born-Oppenheimer approximation: heavy nuclei are treated as fixed classical point charges

From First to Second Quantization

We begin with the electronic Schrödinger equation, whose Hamiltonian is

$$\hat{H}_{\text{elec}} = - \sum_{i=1}^{N_e} \frac{1}{2} \nabla_i^2 - \sum_{i=1}^{N_e} \sum_{A=1}^{N_n} \frac{Z_A}{|\mathbf{r}_{iA}|} + \frac{1}{2} \sum_{\substack{i,j \\ i \neq j}}^{N_e} \frac{1}{|\mathbf{r}_{ij}|} + V_{NN} \quad (24)$$

The corresponding wavefunction

$$\psi(\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_{N_e}) \quad (25)$$

depends continuously on the spatial coordinates of all electrons.

The Problem

This becomes a very high-dimensional partial differential equation, making exact solutions extremely difficult for realistic molecules.

Instead of solving directly for the exact spatial distribution of electrons, quantum chemistry introduces a different viewpoint.

We first choose a finite set of approximate orbitals (using the **LCAO approximation**: Linear Combination of Atomic Orbitals)

$$\{\phi_1, \phi_2, \phi_3, \dots\} \quad (26)$$

and ask:

“What is the probability that electrons occupy these orbitals?”

From First to Second Quantization

The problem therefore changes from

Finding an unknown continuous molecular wavefunction

to

Finding electron occupancies in a discrete orbital basis

This allows us to reformulate the theory using operators acting on a discrete fermionic Hilbert space. The continuous-coordinate wavefunction $\psi(\mathbf{r}_1, \mathbf{r}_2, \dots)$ is expanded in a finite spin-orbital basis $\{\phi_p\}$.

The Hamiltonian is then rewritten using fermionic creation and annihilation operators:

$$\hat{H}_{\text{SQ}} = \sum_{pq} h_{pq} a_p^\dagger a_q + \frac{1}{2} \sum_{pqrs} h_{pqrs} a_p^\dagger a_q^\dagger a_r a_s \quad (27)$$

where

- $a_p^\dagger$: creates an electron in spin orbital p
- a_q : annihilates an electron in spin orbital q
- h_{pq} : one-electron integrals (kinetic energy + electron-nucleus attraction)
- h_{pqrs} : two-electron integrals (electron-electron repulsion)

Continuous Hilbert Space $\rightarrow$ Discrete Fermionic Fock Space

Jordan–Wigner Transformation

After second quantization, the Hamiltonian becomes

$$\hat{H}_{SQ} = \sum_{pq} h_{pq} a_p^\dagger a_q + \frac{1}{2} \sum_{pqrs} h_{pqrs} a_p^\dagger a_q^\dagger a_r a_s \quad (28)$$

- Instead of solving a continuous differential equation, we now work with a discrete operator acting on a fermionic Fock space.
- The Hamiltonian is represented using one-electron and two-electron tensors.
- At worst, the two-electron tensor contains $\mathcal{O}(N^4)$ terms, where N is the number of basis orbitals.
- This representation is significantly more approachable computationally, and forms the foundation of modern computational quantum chemistry.
- However, a quantum computer does not understand fermionic creation and annihilation operators directly.
- We therefore rewrite

$$a_p^\dagger \quad \text{and} \quad a_p \quad (29)$$

using Pauli operators that quantum hardware understands.

- One of the simplest and most intuitive mappings is the **Jordan–Wigner Transformation**.

In the Jordan–Wigner picture:

$$|0\rangle \rightarrow \text{orbital unoccupied} \quad |1\rangle \rightarrow \text{orbital occupied} \quad (30)$$

The fermionic operators are mapped as

$$a_j^\dagger = \frac{1}{2}(X_j - iY_j) \otimes Z_{j-1} \cdots Z_1 Z_0 \quad (31)$$

$$a_j = \frac{1}{2}(X_j + iY_j) \otimes Z_{j-1} \cdots Z_1 Z_0 \quad (32)$$

The VQE Workflow

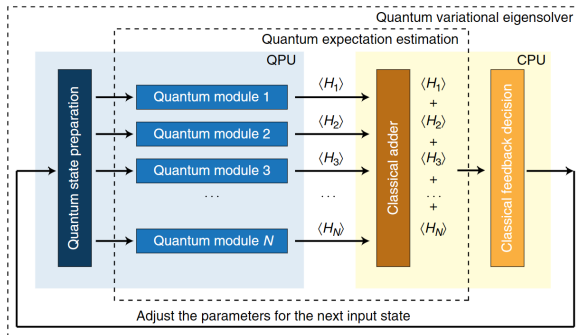


Figure: VQE Workflow. Source : A variational eigenvalue solver on a photonic quantum processor *Nature Communications* volume 5, Article number: 4213 (2014)

Hybrid Nature:

- Quantum computer: state preparation + measurements
- Classical computer: optimization loop

Variational Quantum Eigensolver (VQE) Workflow

- 1 Start with the molecular Hamiltonian in second-quantized form

$$\hat{H}_{\text{SQ}} = \sum_{pq} h_{pq} a_p^\dagger a_q + \frac{1}{2} \sum_{pqrs} h_{pqrs} a_p^\dagger a_q^\dagger a_r a_s \quad (33)$$

- 2 Apply a fermion-to-qubit mapping (Jordan–Wigner / Bravyi–Kitaev)

$$\hat{H}_{\text{SQ}} \rightarrow \hat{H}_Q = \sum_i c_i P_i \quad (34)$$

where P_i are tensor products of Pauli operators.

- 3 Choose a parameterized trial wavefunction (Ansatz)

$$|\psi(\theta)\rangle = U(\theta) |\psi_{\text{HF}}\rangle \quad (35)$$

For chemistry applications, a common choice is the UCCSD Ansatz.

- 4 Prepare $|\psi(\theta)\rangle$ on the quantum computer.
- 5 Measure expectation values of Pauli strings

$$\langle H \rangle_\theta = \sum_i c_i \langle \psi(\theta) | P_i | \psi(\theta) \rangle \quad (36)$$

to estimate the molecular energy.

- 6 Send the measured energy to a classical optimizer.
- 7 The classical optimizer updates the parameters

$$\theta \rightarrow \theta' \quad (37)$$

to minimize the energy expectation value.

Example : H₂ Molecule

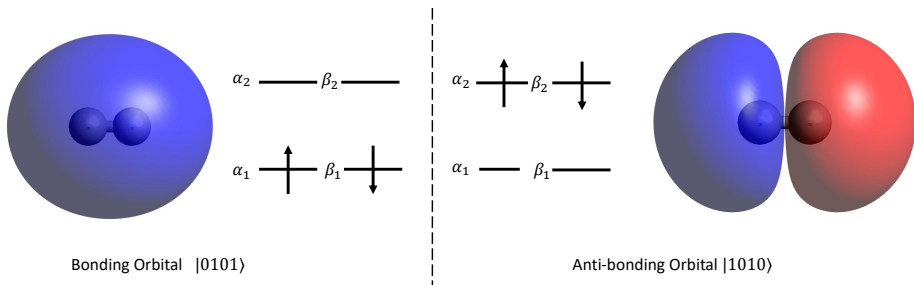


Figure: H₂ molecular orbitals

Hartree-Fock State and Ground State

$$|\psi_{HF}\rangle = |0101\rangle$$

$$|\psi_0\rangle = 0.993647 |0101\rangle + 0.112544 |1010\rangle$$

(39)

Unitary Coupled Cluster Ansatz (UCCSD)

$$|\psi(\theta)\rangle = e^{(T_1+T_2)-(T_1^\dagger+T_2^\dagger)} |\psi_{\text{HF}}\rangle \quad (40)$$

$$T_1 = \sum_{ia} t_i^a a_a^\dagger a_i \quad (\text{single excitations}) \quad (41)$$

$$T_2 = \sum_{ijab} t_{ij}^{ab} a_a^\dagger a_b^\dagger a_j a_i \quad (\text{double excitations}) \quad (42)$$

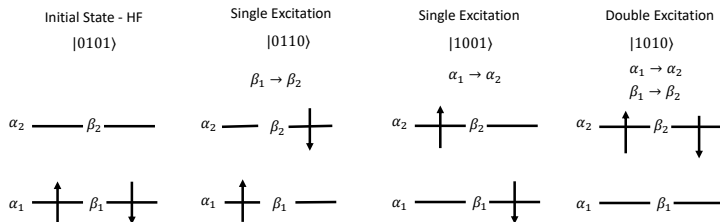


Figure: H_2 molecule - possible excitations

H₂ Molecule - Ansatz Preparation in Qiskit

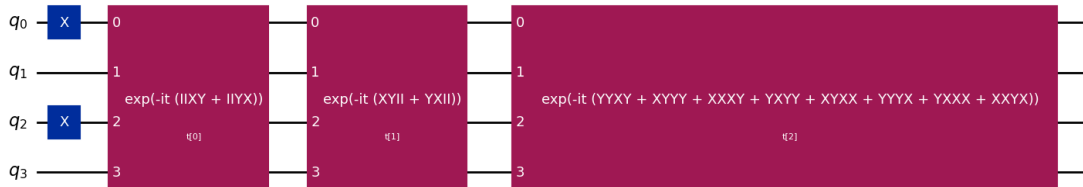


Figure: H₂ VQE implementation with UCCSD Ansatz in Qiskit

Example: H₂ Molecule

Problem: Compute the ground state energy of hydrogen molecule

Workflow

- 1 Perform Hartree-Fock calculation
- 2 Map fermions to qubits
- 3 Build qubit Hamiltonian
- 4 Run VQE optimization

Why H₂?

- Smallest nontrivial quantum chemistry system
- Demonstrates the full quantum workflow
- Often reduced to 2 or 4 qubits

Applications Beyond H₂

- Materials science
- Drug discovery
- Catalysis

Advantages, Challenges, and Outlook

Advantages

- Suitable for NISQ hardware
- Shallow circuits possible
- Flexible ansatz design

Challenges

- Noise and decoherence
- Barren plateaus
- Large measurement overhead
- Optimizer instability

Modern Directions

- ADAPT-VQE
- DMET + VQE
- Error mitigation
- Quantum chemistry at scale

VQE = One of the flagship algorithms of the NISQ era

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